PHP Programming Practice Questions

SECTION 1: BASIC OUTPUT & ECHO/PRINT (5 Questions)

1. Hello World -> Write a PHP script that outputs "Welcome to Web Technology II" using the echo construct.
2. Multiple Echo Parameters -> Write a PHP script that uses echo with multiple parameters (comma-separated) to display: "PHP" "Programming" "is" "Fun"
3. Print vs Echo -> Write a PHP script that demonstrates both echo and print by displaying your name and student ID.
4. HTML Integration -> Write a PHP script that outputs HTML content with PHP, creating a simple webpage with a heading and a paragraph.
5. Using Print in Expression -> Write a PHP script that uses the print construct inside an expression and displays the return value (which is 1).

SECTION 2: VARIABLES & DATA TYPES (10 Questions)

6. Variable Declaration -> Declare variables for your name, age, and height, then display them using echo.
7. Dynamic Typing -> Create a variable named $data. Assign it an integer value, display it, then reassign it to a string value and display it.
8. Variable Variable ($$) -> Create a variable cityName = "Kathmandu", then use a variable to create a new variable named cityName="Kathmandu", then use a variable variable to create a new variable named Kathmandu with the value "Capital City". Display the result.
9. Case Sensitivity -> Create two variables: name and Name with different values. Display both to demonstrate case sensitivity.
10. Integer Operations -> Declare variables for marks in three subjects (Nepali, English, Math). Calculate and display the total and average marks.
11. Float Operations -> Calculate the total cost of 5 items priced at Rs. 199.99 each, including 13% VAT. Display the subtotal, VAT amount, and total.

12. String Concatenation -> Concatenate three string variables: firstName, middleName, lastName and display the full name.
13. Double vs Single Quotes -> Create a variable $college = "Himalayan Darshan". Display it using both double quotes and single quotes to show the difference in parsing.
14. Boolean Conditions -> Create a boolean variable $isStudent = true. Write an if-else statement that displays "Access Granted" if true, otherwise "Access Denied".
15. var_dump Usage -> Use var_dump() to display the type and value of the following variables: age=25, name = "Ram", price = 99.99, isActive = false, $scores = [85, 90, 78].

SECTION 3: CONSTANTS (5 Questions)

16. define() Function -> Define a constant using define() for PI with value 3.14159. Calculate the area of a circle with radius 7.
17. const Keyword -> Define a constant using the const keyword for COLLEGE_NAME and display it.
18. Magic Constants -> Display the following magic constants: LINE, FILE, DIR, FUNCTION inside a function.
19. Constant Case Sensitivity -> Define a constant "GREETING" using define() and try to echo it in lowercase to demonstrate case sensitivity.
20. Global Scope Constant -> Define a constant inside a function using define() and access it outside the function to demonstrate global scope.

SECTION 4: ARITHMETIC OPERATORS (6 Questions)

21. Basic Arithmetic -> Write a PHP script that calculates: (25 + 15) * 4 - 30 / 2. Display the result.
22. Modulo Operator -> Write a script that checks if a number is even or odd using the modulo operator (%).
23. Exponentiation -> Calculate the compound interest using the formula: A = P(1 + r)^n. Use P=10000, r=0.08, n=3.

24. Temperature Conversion -> Convert Celsius to Fahrenheit using the formula: F = (C * 9/5) + 32. Test with 37°C.
25. Area Calculation -> Calculate the area of a rectangle (length=15, width=8) and display "The area is [result] square units".
26. BMI Calculator -> Calculate BMI using: weight (kg) / (height in meters)^2. Use weight=70, height=1.75.

SECTION 5: ASSIGNMENT OPERATORS (5 Questions)

27. Compound Assignment -> Start with $x = 10. Use compound assignment operators (+=, -=, *=, /=, %=) to modify and display x after each operation.
28. Increment/Decrement -> Demonstrate pre-increment and post-increment operators. Show how a++ and ++a differ.
29. Assignment Chain -> Chain assignments: $a = $b = $c = 50. Display all three variables to verify they have the same value.
30. Swap Values -> Swap the values of two variables (a=10, b=20) without using a third variable.
31. Shopping Discount -> Calculate the final price after applying a 15% discount to an item costing Rs. 4500. Use assignment operators in the calculation.

SECTION 6: COMPARISON OPERATORS (5 Questions)

32. Equality vs Identity -> Compare 5 == "5" and 5 === "5". Display the results and explain the difference.
33. Age Check -> Check if a person aged 18 is eligible to vote. Display "Eligible" if true, otherwise "Not Eligible".
34. Comparison Chain -> Compare three numbers (a=45, b=60, c=45) using various comparison operators and display boolean results.
35. Password Validator -> Check if a password length is greater than or equal to 8 characters. Test with "secure123".

36. Grade Checker -> Check if a student's score (75) is greater than or equal to 40 and less than or equal to 100.

SECTION 7: LOGICAL OPERATORS (5 Questions)

37. AND Operator -> Check if a number (25) is between 10 and 50 using the AND (&&) operator.
38. OR Operator -> Check if a year (2024) is divisible by 4 OR by 400 to determine if it's a leap year.
39. NOT Operator -> Use the NOT (!) operator to check if a variable $isLoggedIn = false. Display "Please Login" if not logged in.
40. Combination Operators -> Check if a student has passed both Math (score=75) AND English (score=65) with passing marks of 40 each.
41. Login System -> Create a simple login check: username is "admin" AND password is "1234". Display "Welcome Admin" if true, otherwise "Invalid Credentials".

SECTION 8: STRING OPERATORS (4 Questions)

42. Concatenation Operator -> Combine firstName="Sita" and lastName = "Sharma" using the concatenation operator (.).
43. Concatenation Assignment -> Start with $message = "Hello". Use .= to append " World!" and display the result.
44. Building HTML -> Use string concatenation to build an HTML table row with columns: Name, Age, City.
45. Email Generation -> Generate an email address by concatenating firstName, adot(.), lastName, and "@example.com".

SECTION 9: CONDITIONAL OPERATORS (4 Questions)

46. Ternary Operator -> Use the ternary operator to check if a number (15) is even or odd and display the result.

47. Null Coalescing -> Use the null coalescing operator (??) to assign a default value if $username is null.
48. Nested Ternary -> Use nested ternary operators to display "Excellent", "Good", or "Needs Improvement" based on a score of 85.
49. Discount with Ternary -> Apply a 20% discount if the total bill is greater than Rs. 5000, otherwise apply a 5% discount. Use ternary operator.

SECTION 10: CONTROL STATEMENTS (10 Questions)

50. if-else Statement -> Create a script that checks if a number is positive, negative, or zero. Display appropriate messages.
51. else-if Ladder -> Convert a numerical grade (0-100) to a letter grade: A (80-100), B (60-79), C (40-59), D (20-39), E (0-19).
52. switch Statement -> Use switch to display the day name based on a number (1-7). Monday=1, Tuesday=2, etc.
53. switch with Default -> Create a menu system using switch: Option 1 (Add), Option 2 (Edit), Option 3 (Delete), Default (Invalid Option).
54. match Expression (PHP 8+) -> Use match expression to assign a status message based on HTTP status codes: 200="OK", 404="Not Found", 500="Server Error".
55. for Loop -> Write a for loop that prints numbers from 1 to 10.
56. for Loop with Sum -> Calculate the sum of all numbers from 1 to 100 using a for loop.
57. while Loop -> Use a while loop to print even numbers from 2 to 20.
58. do-while Loop -> Use a do-while loop to display numbers from 10 down to 1.
59. foreach Loop -> Create an array of five city names and use foreach to display each city with its index.

SECTION 11: FUNCTIONS (8 Questions)

60. Function Definition -> Create a simple function called displayGreeting() that echoes "Welcome to PHP Programming!".

61. Function with Parameters -> Create a function called calculateArea(length,length,width) that calculates and returns the area of a rectangle.
62. Return Value -> Write a function called squareNumber($num) that returns the square of the number.
63. Default Parameters -> Create a function called generateEmail(firstName, lastName, $domain = "gmail.com") that returns a complete email address.
64. Multiple Parameters -> Write a function calculateDiscount(price, discountPercent) that returns the discounted price.
65. Function with Array -> Write a function that accepts an array of numbers and returns their sum and average.
66. Scope Demonstration -> Create a function that attempts to access a variable defined outside the function. Show that it's inaccessible.
67. Type Hinting -> Write a function with strict typing that adds two integers and returns the result. Test with integer and string inputs.

SECTION 12: FILE INCLUSION (3 Questions)

68. include_once -> Create a file named "config.php" containing database configuration constants. Include it in another script using include_once.
69. require vs include -> Demonstrate the difference between require and include by including a missing file in both ways.
70. Modular Header/Footer -> Create separate header.php and footer.php files and include them in a main page to create a complete HTML layout.
